

$$\begin{bmatrix} 1 & k \\ k & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Augmented matrix:

$$\left[\begin{array}{cc|c} 1 & k & 1 \\ k & 1 & -1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - kR_1$$

$$\left[\begin{array}{cc|c} 1 & k & 1 \\ 0 & 1-k^2 & -1-k \end{array} \right] \rightarrow \begin{aligned} (1-k^2)y &= -(1+k) \\ \hookrightarrow (1+k)(1-k)y &= -(1+k) \end{aligned}$$

① Unique solⁿ:

$$\rightarrow k \neq -1, k \neq 1$$

$$\hookrightarrow y = \frac{-1}{1-k}, x = \frac{1}{1-k}$$

~~Rank(A) = 2~~

$$\text{Rank}(A) = 2 \quad \& \quad (n=2) \\ \text{rank}[A|b] = 2$$

② No solⁿ:

$$k = 1$$

$$\text{Row 2 becomes } \rightarrow (0 \quad 0 \quad | \quad -2) \quad \begin{aligned} \text{rank}(A) &= 1 \quad \& \\ \text{rank}[A|b] &= 2 \end{aligned}$$

$$(0 = -2) \rightarrow \text{false}$$

so no solⁿ

Ranks differ — no solⁿ

③ ∞ solⁿs:

$$k = -1$$

$$\text{Row 2} \rightarrow (0, 0, 0).$$

(says $0 = 0$).

Only 1 eqⁿ remaining: $x - y = 1$

so ∞ solⁿs.

$$\text{rank}(A) = 1 = \text{rank}[A|b]$$

Ranks are equal but less than 2 so ∞ solⁿs